

Configuration Reference

Configuration is a YAML file (default: config/default_config.yaml). Override it with `python launcher.py --config /path/to/your.yaml`. Values not present in your file fall back to the defaults below. Runtime edits are possible via `PUT /api/v1/config` (ENGINEER role) after validation.

The JWT secret is **never** in this file. It comes only from the `AIRONE_JWT_SECRET` environment variable.

api

Key	Default	Effect
host	127.0.0.1	Bind address for the REST API (loopback; <code>AIRONE_API_HOST</code> overrides; 0.0.0.0 logs a warning — use a TLS proxy).
port	5000	API port.
max_body_bytes	262144	Maximum request body; larger → 413.
cors_origins	[http://localhost:3000, http://127.0.0.1:3000]	Exact allowed browser origins. "*" refused unless <code>AIRONE_ALLOW_CORS_WILDCARD=1</code> .
rate_limits.auth_login	10 per minute	Login rate limit.
rate_limits.auth_refresh	30 per minute	Token refresh limit.
rate_limits.telemetry_export	5 per minute	Export limit.
rate_limits.ml_training	2 per hour	Training limit.
rate_limits.config	20 per minute	Config change limit.

security

Key	Default	Effect
access_token_minutes	15	Access-token lifetime (hard cap 1440). Alternative key <code>jwt_expiry_minutes</code> also accepted.

Key	Default	Effect
refresh_token_days	7	Refresh-token lifetime (hard cap 30). Alternative key refresh_expiry_days also accepted.
bcrypt_rounds	12	Password hashing cost.
max_login_attempts	5	Per-account failures (15 min window) before logout.
max_login_attempts_per_ip	20	Per-source-IP failures before logout.
lockout_seconds	300	Initial logout; doubles on repeated lockouts (cap 1 h).

Secrets never live in this file: AIRONE_JWT_SECRET (required), AIRONE_ADMIN_PASSWORD (bootstrap only), AIRONE_LINK_KEY (optional).

telemetry

Key	Default	Effect
serial_port	null	Serial device; null = auto-discover.
baud_rate	115200	Serial baud rate.
buffer_size	65536	Receive buffer size (bytes).
clock_drift_tolerance_ms	5000	Max tolerated clock drift before a timestamp is flagged.
require_authenticated_frames	false	true requires AIRONE_LINK_KEY and marks every frame without a valid HMAC tag INVALID.

pipeline

Key	Default	Effect
enabled_stages	null	List of stage numbers to run; null = all 20.
required_fields	[]	Fields whose absence marks a frame incomplete.
filter_config.fused_pressure	Kalman 1-D (process_noise 0.001, measurement_noise	Pressure filter.

Key	Default	Effect
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filter_config.temperature	moving average (window 5)	Temperature filter.
filter_config.altitude	EKF (process_noise 0.01, measurement_noise 1.0)	Altitude filter.

storage

Key	Default	Effect
db_path	data/airone.db	SQLite database path.
data_dir	data	Data directory (exports, snapshots, models).
backup_interval_hours	6	DB backup cadence.
max_db_size_mb	2048	Soft size ceiling.

mission

Key	Default	Effect
default_primary	atmospheric_profiling	Primary objective.
default_secondary	uv_radiation_mapping	Secondary objective.

logging

Key	Default	Effect
level	INFO	Log level.
log_dir	logs	Log directory.
json_output	true	Structured JSON logs.

ml (optional block)

Not present in the default file; add it to tune the ML worker.

Key	Default	Effect
window_size	240	Rolling window (frames) for live anomaly scoring.
run_interval_s	5.0	Seconds between scoring passes.
min_samples	20	Minimum valid samples before scoring/training.

Key	Default	Effect
auto_fit	true	Auto-fit the ensemble on the current window.
model_dir	data/models	Where checksummed model artefacts are stored.
features	default 6-feature set	Feature columns to extract.

Other top-level keys

Key	Default	Effect
mission_id	default	Mission id used for storage/queries.

admin_password / viewer_password are **no longer read**: accounts are created with launcher.py --create-admin / --create-user (see [security_model.md](#)).

Runtime-mutable keys

PUT /api/v1/config accepts only mission_id, objectives, default_primary, default_secondary, pipeline, scientific, ml, gui, simulation, logging. api, security, storage and telemetry are refused with 400 — change them in the file and restart.